5350 **Advanced Business Analytics**

# Module 6 - MLOps – What it is , Why MLOps

Ivan Portilla
Jesus.Portilla@Colorado.edu
Portilla@gmail.com
github.com/jiportilla/giveback

IBM

# Objectives of This Module

Upon completion of this module, you will understand:

**MLOps - What & Why**

- Definition & People of MLOps - https://ml-ops.org/
- Key MLOps Features
  - Model Development
  - Monitoring
  - Productionalization & Deployment
  - Iteration & Lifecycle
  - Governance
- Lab: Intro to MLOps

# Fall 2024
# 5350 Applied Deep Learning

- This course will give students the language, knowledge, and actionable methods to work alongside technical and non-technical members of your team to create AI solutions.

- Students will explore what it means to design artificial intelligence systems as a team, guided by a clear intent and a focus on people. This course will give you the framework and tools you need to recognize responsible AI design, align your team, and work with data sources to start building AI solutions.

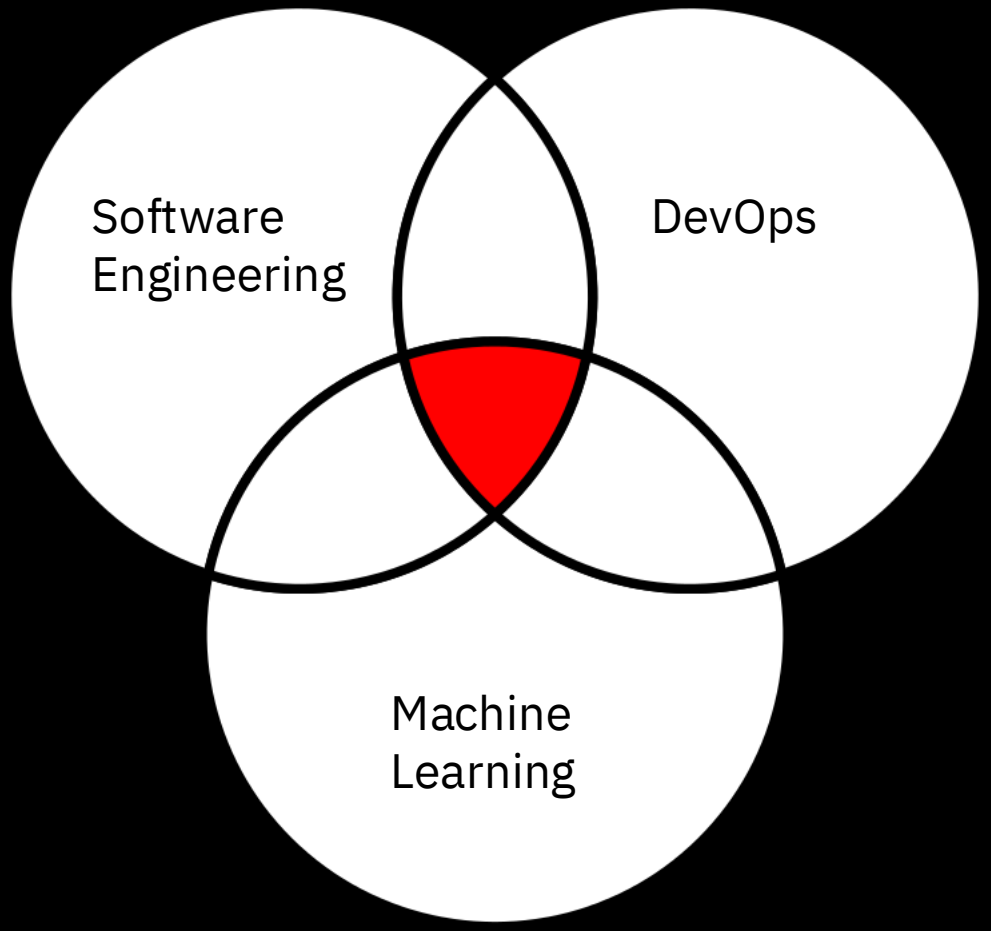
- Students will learn the tools, technology, and practices that enable cross-functional AI teams to efficiently deploy, monitor, retrain, and govern models in production systems.

# Agenda

MLOps What & Why

1. MLOps definition
2. MLOps principles
3. MLOPs Lifecycle
4. AI Roles
5. Operationalize AI
6. Best practices

# MLOps is ...



# MLOps is ... Software Engineering

How do we write software:

From punch cards to multi-user UNIX systems...

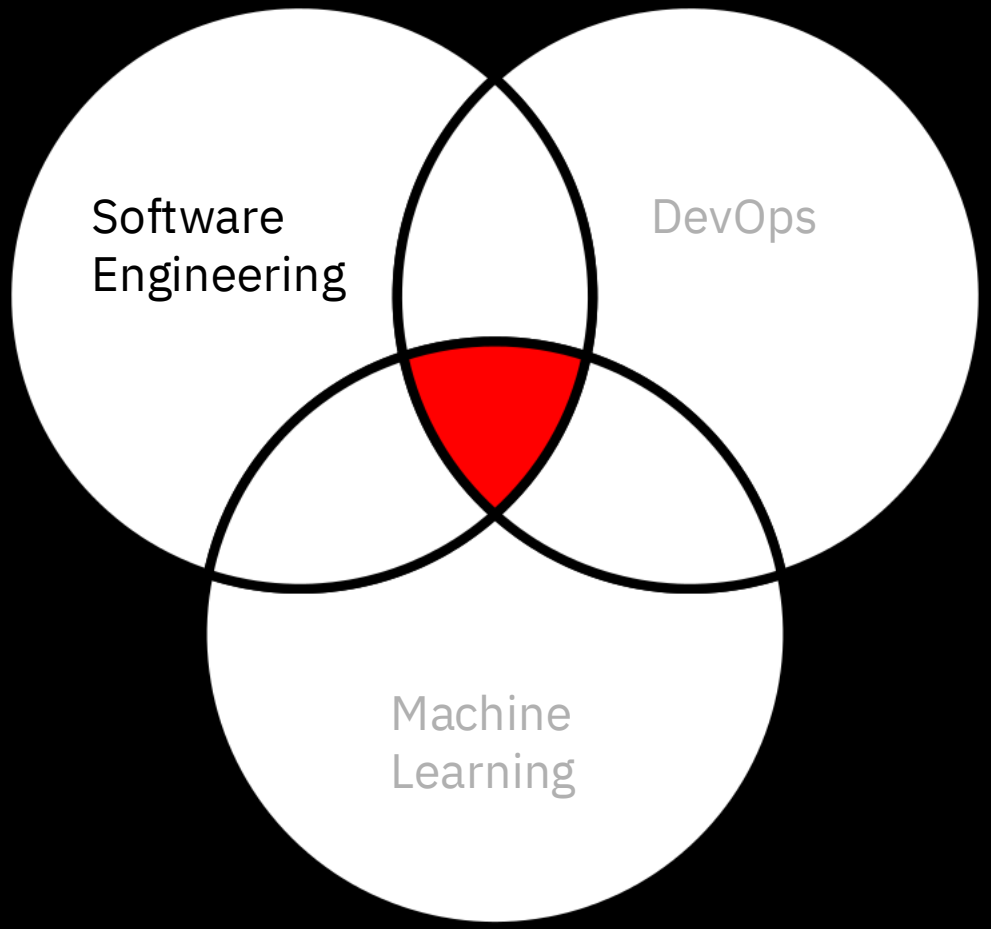
To personal computers...

To networked computers...

Emailing patch files...

Centralized version control...

Distributed version control: Git, Github



# MLOps is ... DevOps

How do we deploy software:

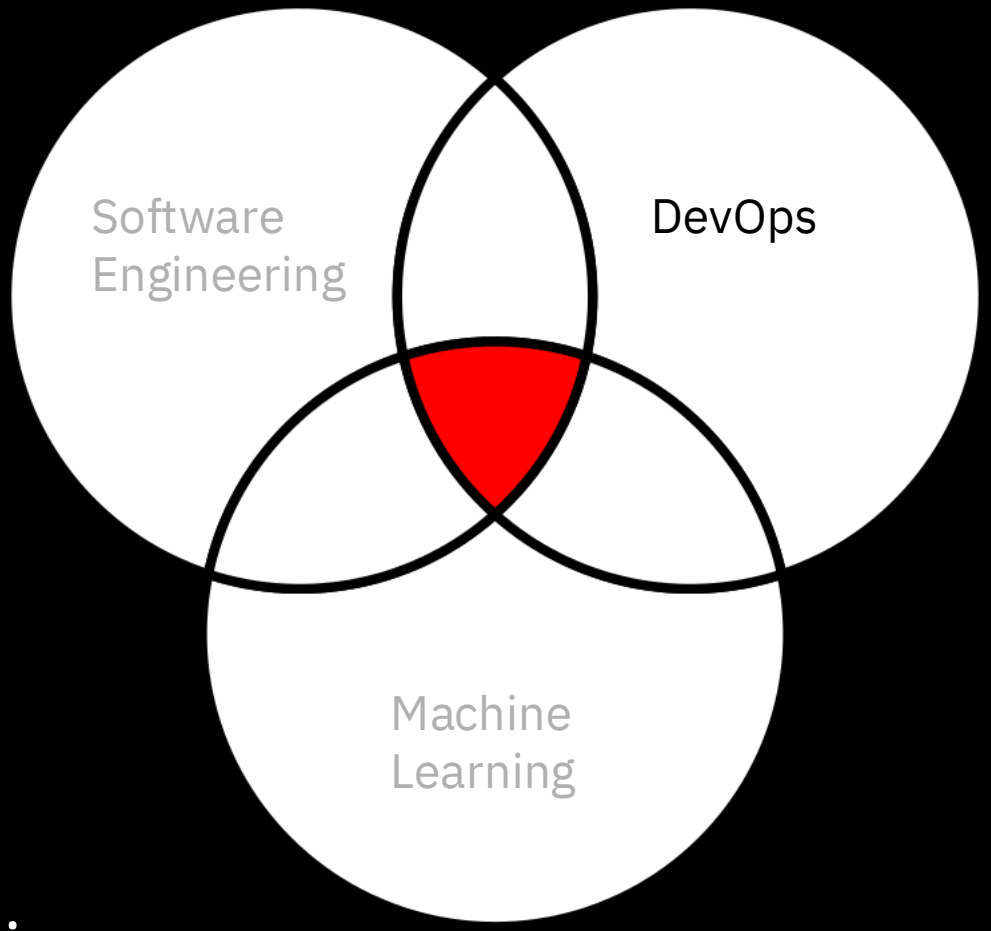
From editing code live on the server...

To building binaries & email them around...

To continuous integration...

To public cloud...

To immutable infrastructure...
Docker, Kubernetes, GitOps



# MLOps is ... Machine Learning

How do we use data & math to train models:

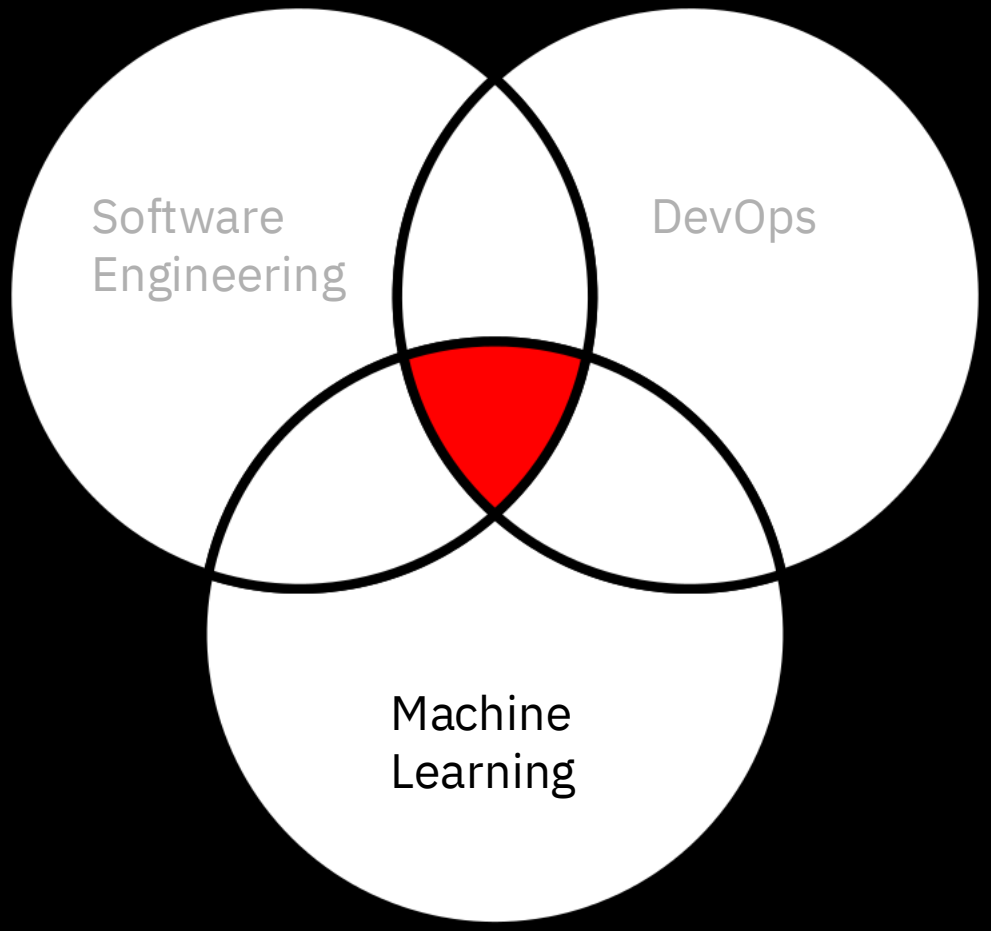
From curve fitting, linear regressions...

To Markov chains ... to neural networks

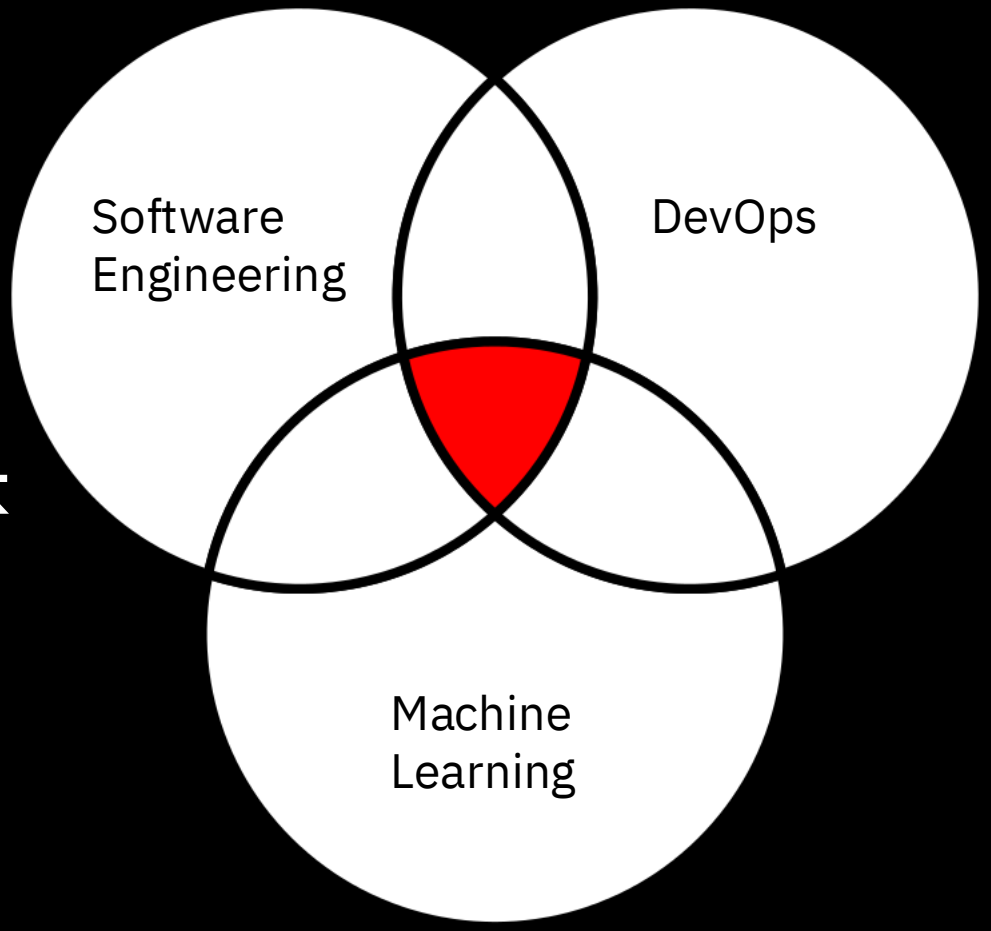
To the first AI winter...

Rediscovery of backdrop, move to data driven approach...

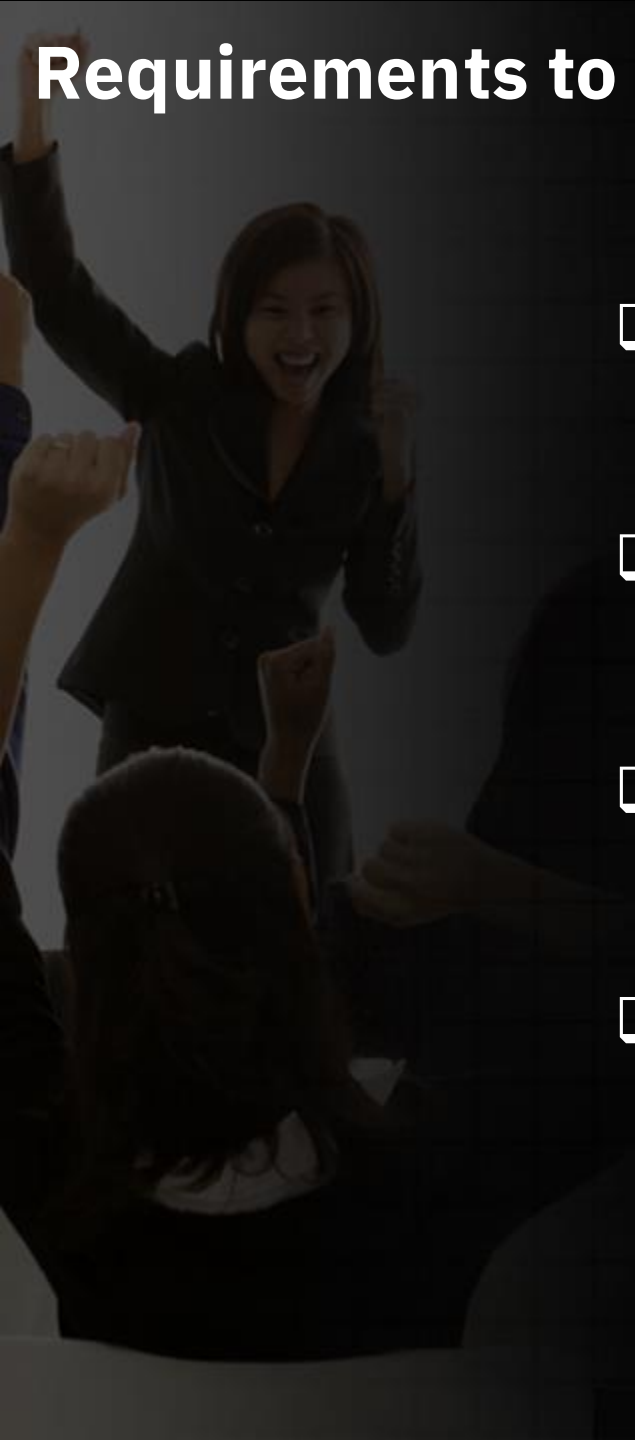
Deep learning becomes computationally feasible



**MLOps** is the **convergence** of Software Engineering, Machine learning & DevOps



# Requirements to Achieve MLOps



- ❑ Reproducible

  Must be able to re-train a 9-month-old model to within few %

- ❑ Accountable

  Must be able to track back from model in Production to its provenance

- ❑ Collaborative

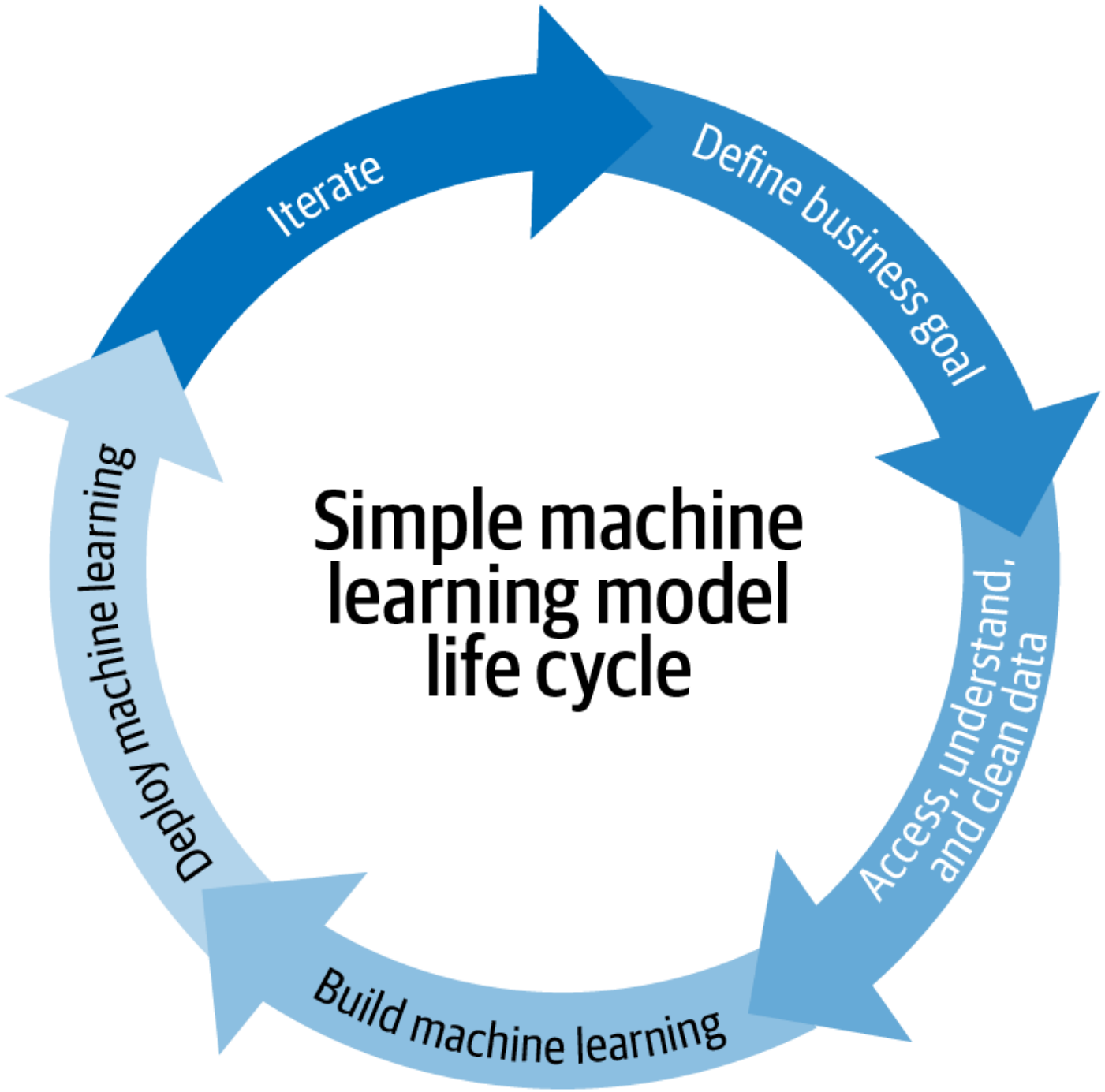
  Must be able to do asynchronous collaboration

- ❑ Continuous

  Must be able to deploy automatically & monitor statistically

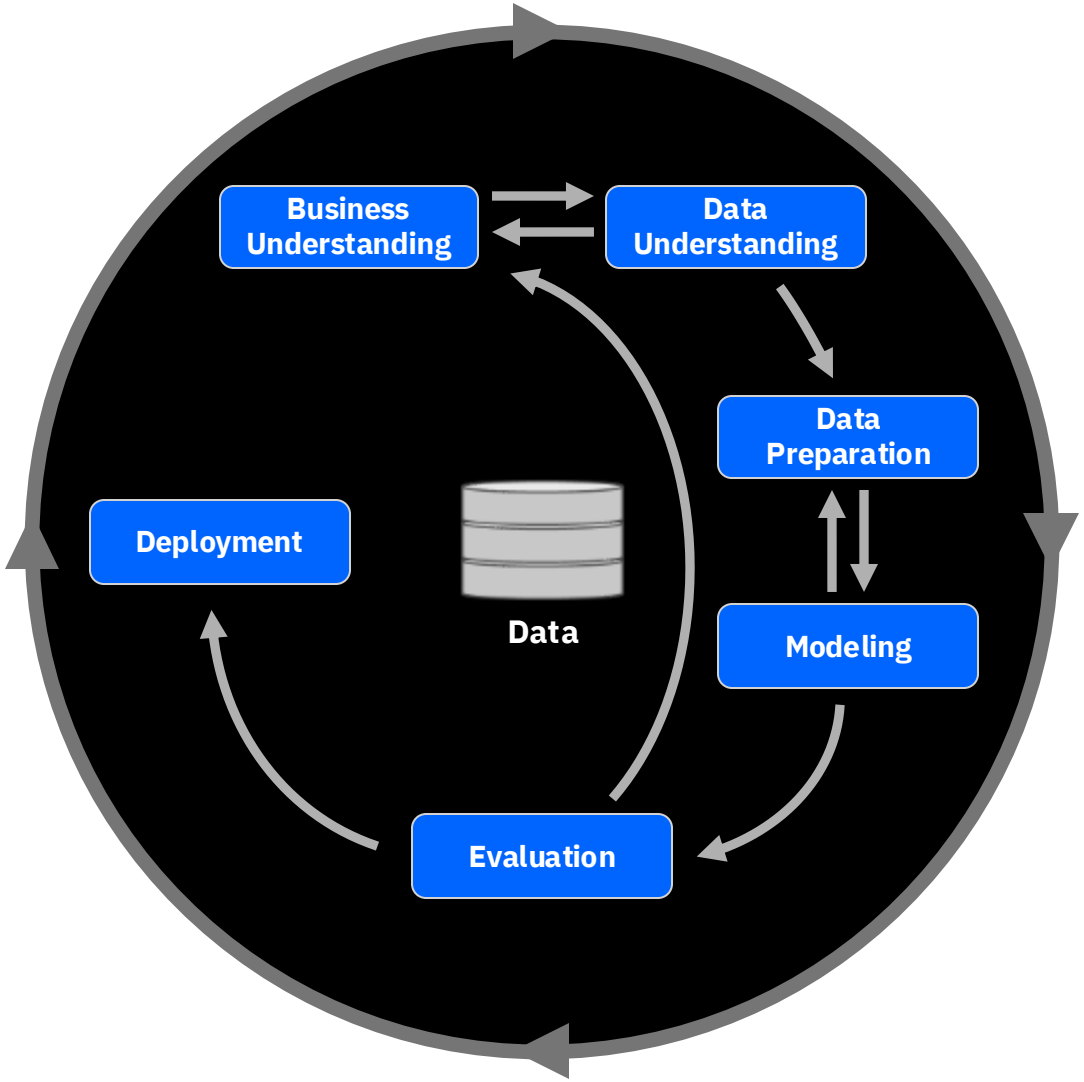
https://mlops.community

# ML Lifecycle



# Solution Development Method Approach

CRossIndustry Standard Process for Data Mining (CRISP-DM)




## Seven steps to successful Data Mining/Predictive Analytics

1. Define the business challenge in a precise statement
2. Define the data model and data requirements
3. Source data from all available repositories
4. Evaluate the data quality
5. Select the machine learning algorithm
6. Interpret the results and iterate to improve model
7. Deploy the model into your business

# AI Project Roles

ANALYST, AND ALEX, THE APP

MaxDome

Drives governance policy effectiveness while tracking how data is used and its value to the company

## Data Steward

Builds data pipelines that power dashboards and data platforms while ensuring high quality

## Data Engineer

Prepares data to tease out the insights they're looking for, without IT involvement

## Data Scientist

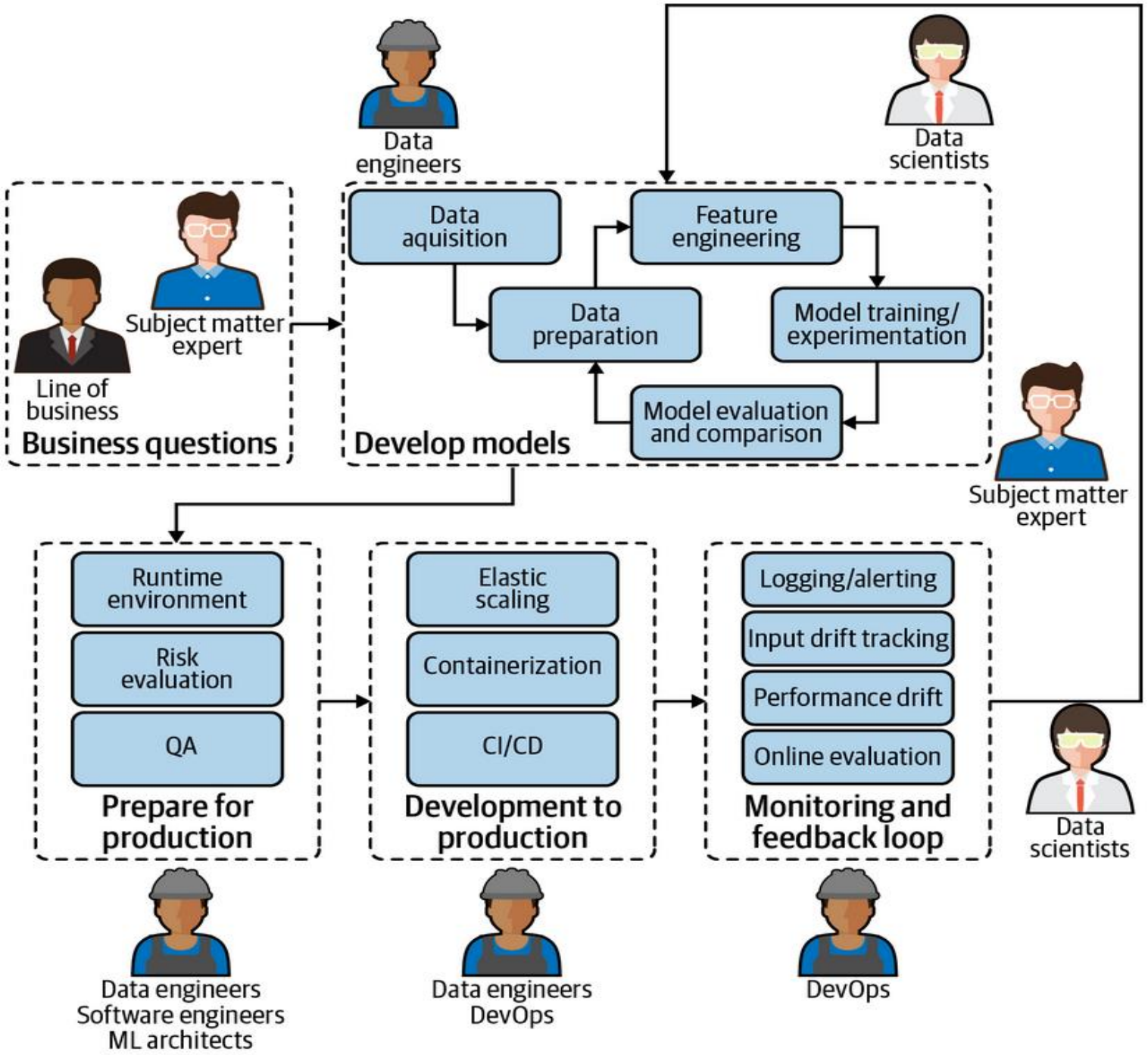
## Business Analyst

Works with data to apply insights to the business strategy

## App Developer

Makes insights immediately actionable and adds intelligence to apps in straightforward manner

## Enterprise ML Lifecycle



# Use case

1. Operationalized AI stages
2. Best practices

CASE STUDY

# IBM helps Walmart set up an enterprise-wide framework for **Trusted AI**

CHALLENGE

Help Walmart ensure fairness and trust is established for their AI/ML models, to accelerate efforts around digital citizenship and social responsibility.

SOLUTION

Platform:
OpenScale (Cloud Pak for Data) on Azure, GCP

Process:
MLOps-Trust (part of AI Factory)

People:
DSE & Expert Labs teams

*Quickly* and *consistently* **detect and counteract bias in Walmart's AI/ML models** and support business decisions.

IBM's Watson OpenScale on Cloud Pak for Data monitors health and fairness in Walmart's hiring system. This enables transparency in the decision making, both for **uniformity of decisions** and to **"explain" any specific decision.** The monitoring and explanations allow Walmart to inspect, understand, and correct decisions, maintaining its commitments as a fair employer, and fairness for each of its candidates and new employees.

Walmart is now expanding beyond the hiring use case to an enterprise-wide **Trusted AI** paradigm for all AI/ML models

# Operationalizing AI involves challenges across personas

C-Level Exec

*"We want to get down from 6-9 months for building & deploying AI to 6-8 weeks. Being able to get to market quickly is very important"*

## Operational Challenges

Data Science Lead

*"Our teams use a variety of ML tools; we need consistent model management"*

Ops Team Lead

*"Need to integrate with our Continuous Integration and Continuous Delivery (CI/CD) process"*

Risk Team Lead

*"How do we consistently trust what the models recommend?"*

## Business View

Business Executive

*"How do we apply AI to transform our business?"*

*"How do we go from ideas to market quickly?"*

## Approach

C-Level Exec

*"We need a strategy and repeatable process for AI at scale and at speed"*

# Creating an AI/ML model for a proof-of-concept looks like this…

Build AI assets

Connect data

Analyze data

Prepare data

Build models

Train models

Visualize & evaluate models

Test for Bias, Fairness

Data engineer

Data scientist

But this is not the complete AI lifecycle!

# Stages in Operationalizing AI

Continuous integration / continuous delivery pipeline

## Scope AI project(s)

Explore, prioritize, select use cases – feasibility vs impact

Detail selected use case(s) – KPIs, data, workflow, success criteria

Technical design: tools, infrastructure, data details, approach

Business

Data Scientist

IT/ Ops

Data provider

## DataOps

All: Define sources & needs

Provider: Set up catalog for discovery, lineage, curation, access rules

Steward: Create data policies & access

Consumer: Find, understand, add, explore, review, share

Data provider

Data steward

## Build AI assets

Connect data

Analyze data

Prepare data

Build models

Train models

Visualize & evaluate models

Test for Bias, Fairness

Data engineer

Data scientist

## ML Ops – Deploy

Review, 3rd party oversight, unit tests, validation, approve deployable version

Deploy model to ML runtime engine

Monitor & evaluate model execution

Manage against thresholds

Validation reports

Software engineer

IT/ Ops

Data scientist

## Manage and Trust

Configure monitoring
Configure integration with other systems

Monitor and manage:

1. Quality
2. Performance
3. Custom metrics
4. Bias/fairness
5. Accuracy & Data Drift

Explain on demand

Risk & Compliance

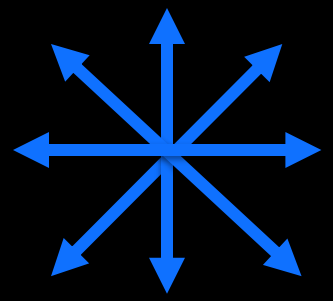
Business

# Trustworthiness presents new challenges to operationalizing AI

**Bias**

Training data and AI models may be biased.

Are privileged groups at a systematic advantage compared to unprivileged groups?



**Quality**

Models need to perform well across the AI/ML lifecycle.

Are relevant performance metrics being monitored over time?

**Drift**

Changes in input data cause model to make inaccurate decisions.

Training data may not include data ranges or combinations seen in real life

**Explainability**

Traditional statistical models are simpler to interpret and explain.

At what point would the outcome have been different?

# Process: Best Practices Throughout the Life Cycle

**Scope**

- Strategic planning provides the foundation
- Structured design thinking defines business need
- Structured exercises translate business need into specific technical task and shape understanding steps

**Understand**

- Flesh out details of contributing data sources according to model
- Flesh out details of technical task and approach based on experiments with clear validation criteria

**Build**

- Develop in agile sprints
- Continuously validate and adapt approach, data, and integration plan
- Develop for data science outcome *and* for integration into broader systems

**Deploy**

- Integrate into software applications and business workflow
- Automate, integrate with CI/CD systems
- Deploy based on QA of both data science outcomes and full system

**Trust**

- Monitor for accuracy and drift to identify the need to update the model(s)
- Demystify AI models by explaining key results
- Identify and correct for unfair outcomes

Continuous Improvement

# Agenda

MLOps What & Why

1. ML Operationalization stages & personas
2. ML operationalization environments
3. MLOps Frameworks
4. MLOps in Action

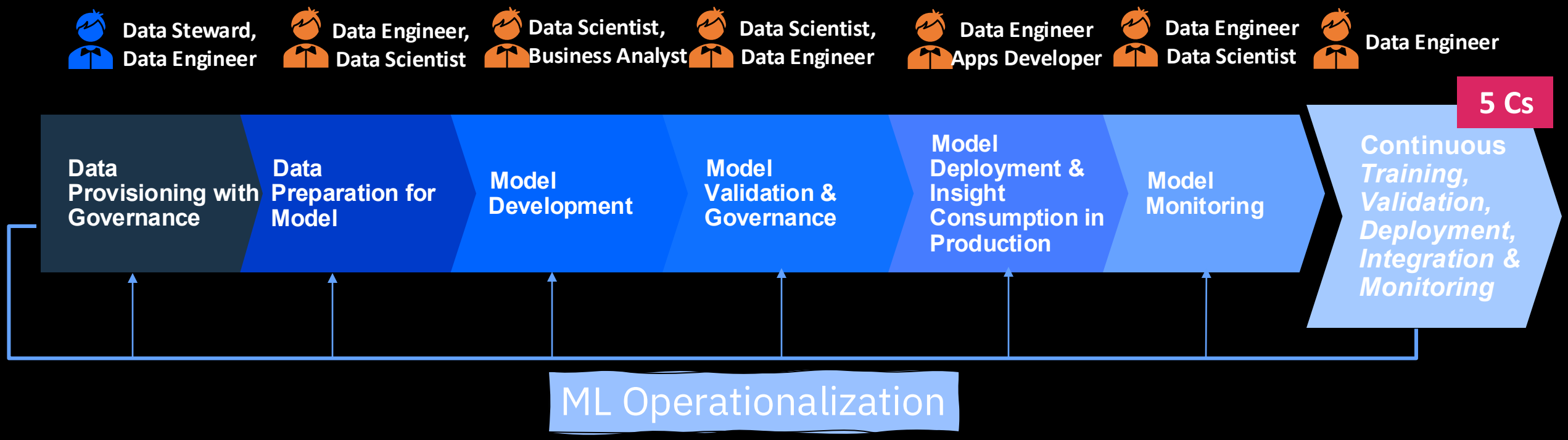
# AI Factory

1. Case study
2. Operationalizing AI stages
3. Best practices

# ML Operationalization – High Level Steps and Personas

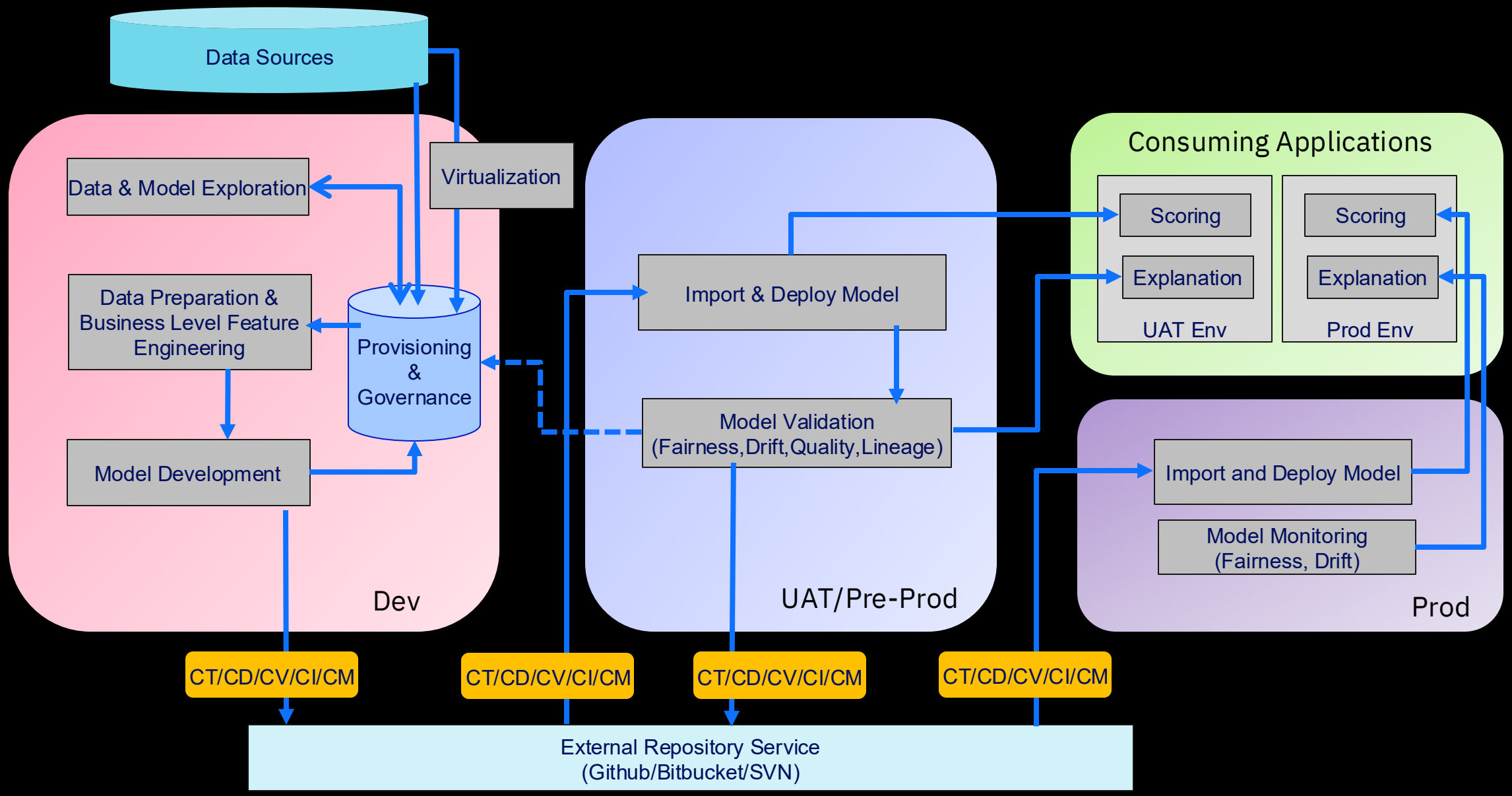
***ML Operationalization refers to operationalization of Machine Learning Models for production use to realize business value out of those Models.***

***ML Operationalization overlays paradigm of DevOps on Model Lifecycle management process (CRISP-DM)***




For Conceptual View of ML Ops please check - https://ibm.co/AI-Ops

# ML Operationalization spread across Dev, UAT & Prod Environments



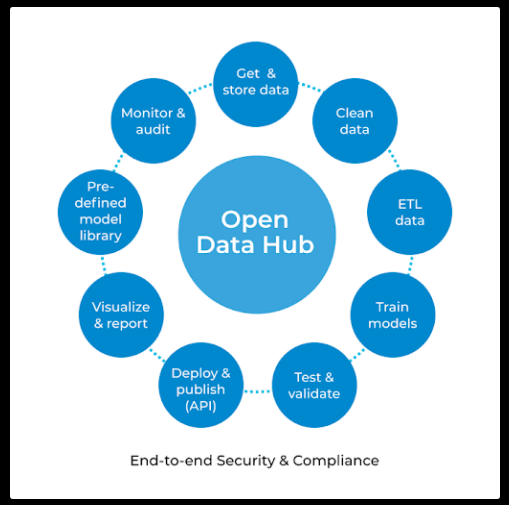
# What to look for in ML Frameworks ?

| Features | Description |
|---|---|
| **Flexibility/Customizability** | How flexible is the platform in integrating and/or customizing new frameworks for AI model development. |
| **Ease of Use** | How easy is it to leverage these tools and proposed techniques from setup to application. |
| **Integrations** | How well does the platform integrate with Git or other model versioning and source control tools, catalogs (for governance and discoverability) or various data sources. |
| **Governance** | How well does the solution support governance and discoverability of assets (data assets, models, notebooks, ...) |
| **Platform** | Support for various platforms (public cloud, on-prem, hybrid cloud), and compute types (CPU/GPU) for training and scoring (or inference) AI models |
| **Monitoring** | How well does the solution support monitoring AI models for performance / explainability / fairness |
| **Scalability** | How scalable is the platform in supporting various Data & AI users in different roles to explore, develop, and deploy AI models. |
| **Openness** | How well does the platform support open-source technologies which has become a key differentiator for platform providers. |
| **Security** | How well does the platform support enterprise-grade security access to the platform in terms of authorization and authentication |
| **Support for 5 Cs** | Support for Continuous Training, Validation, Deployment, Integration and Monitoring |

# Popular ML Ops Tools and Frameworks

## Open Source Frameworks

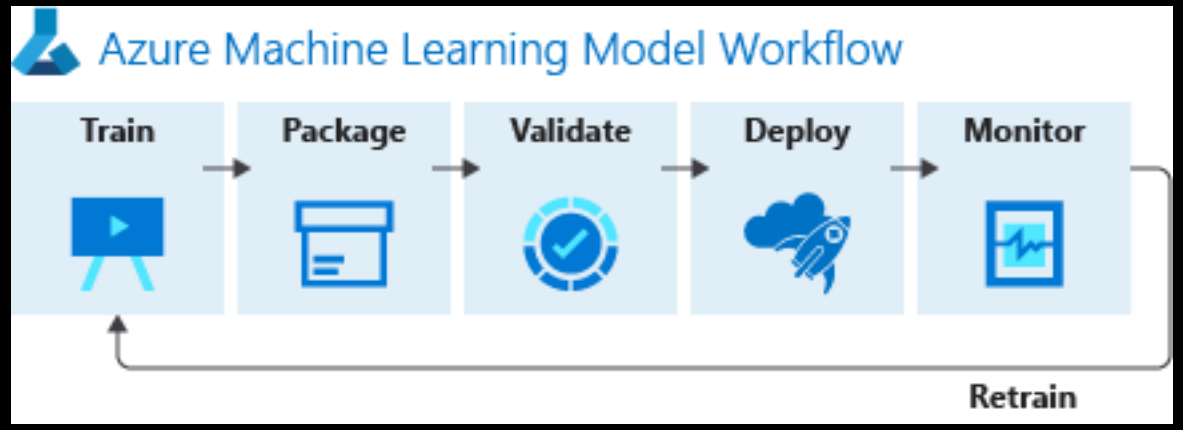
Kubeflow



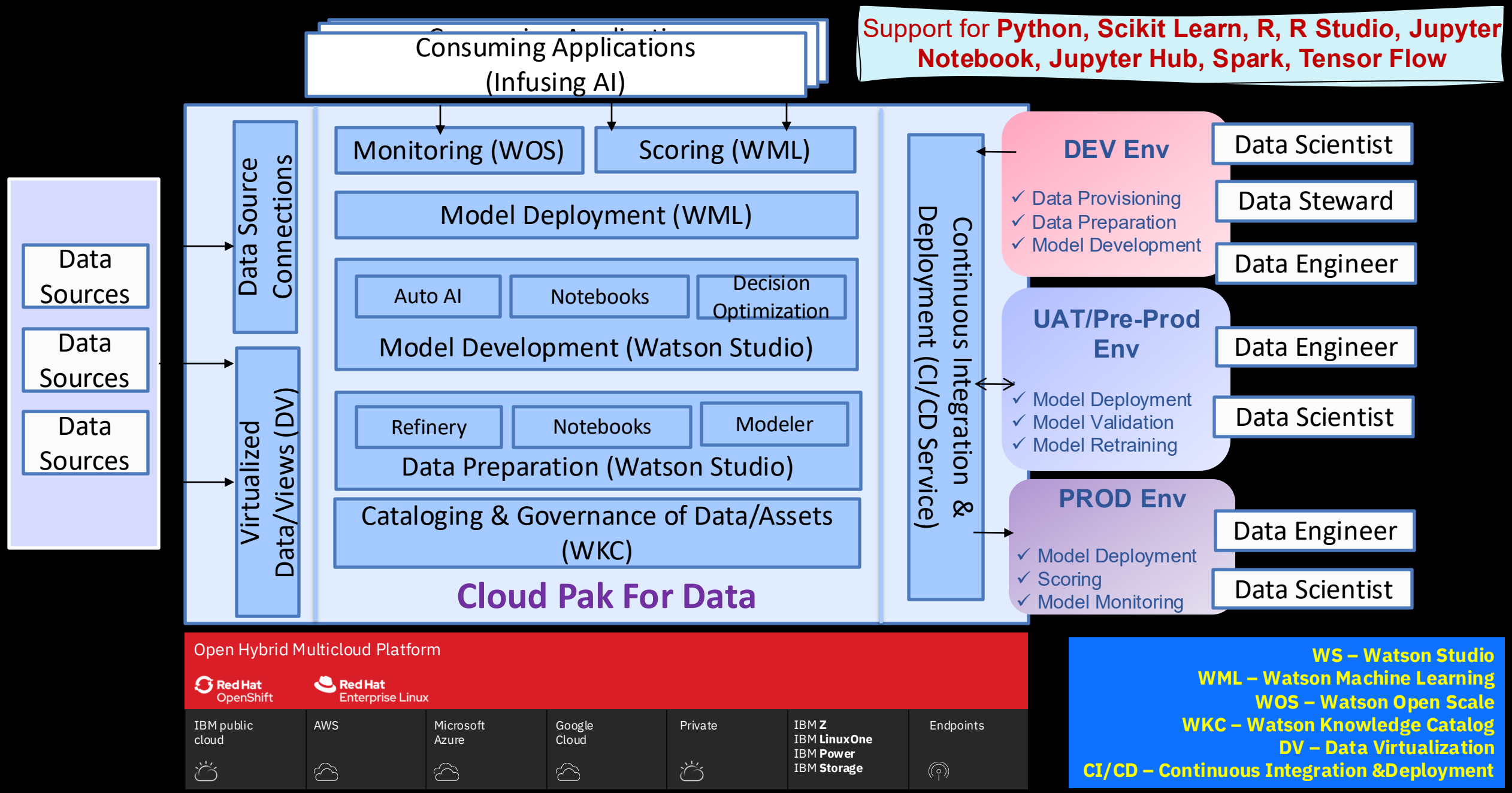

mlflow

Apache Airflow

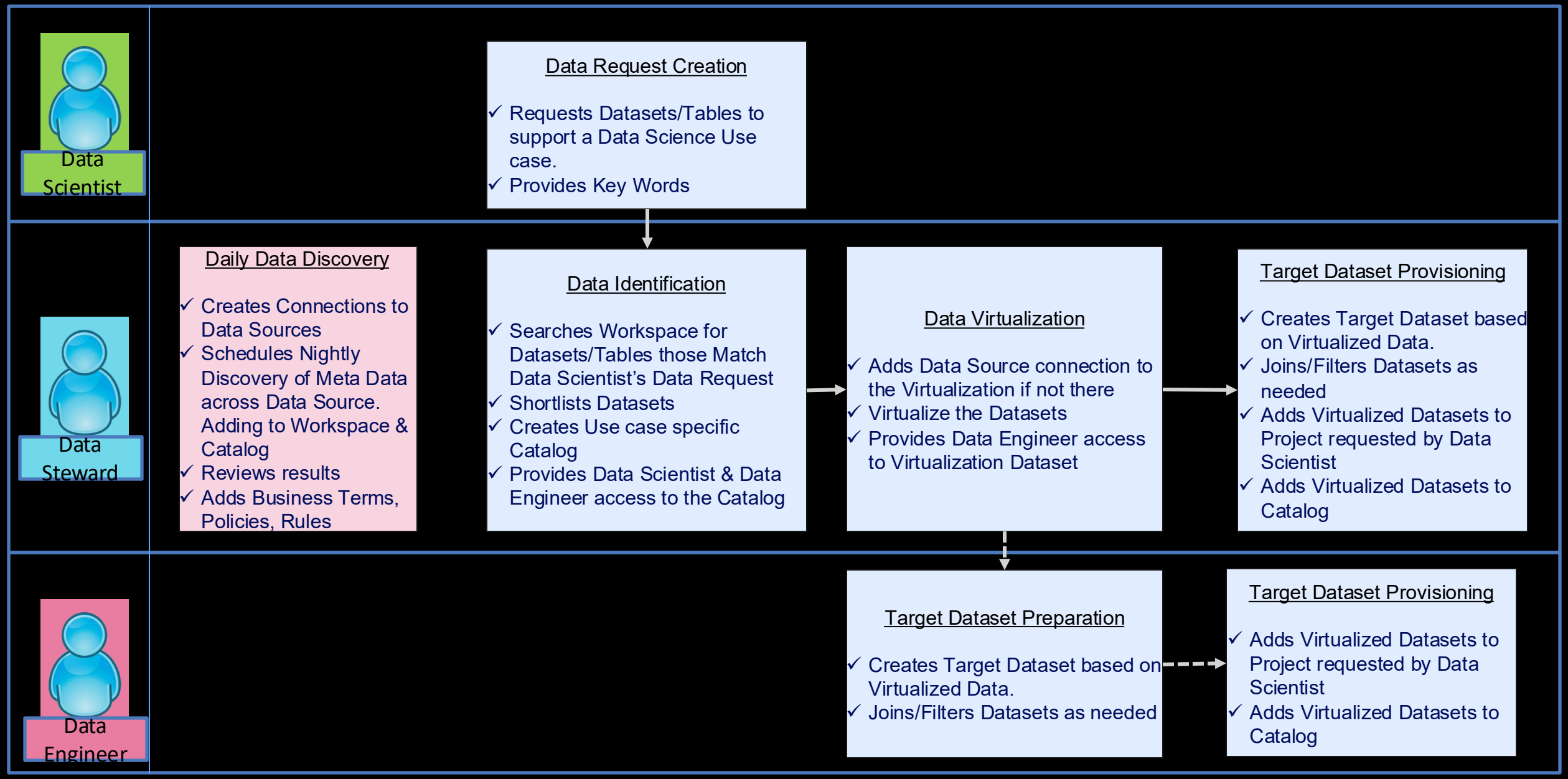
## From Software Vendors



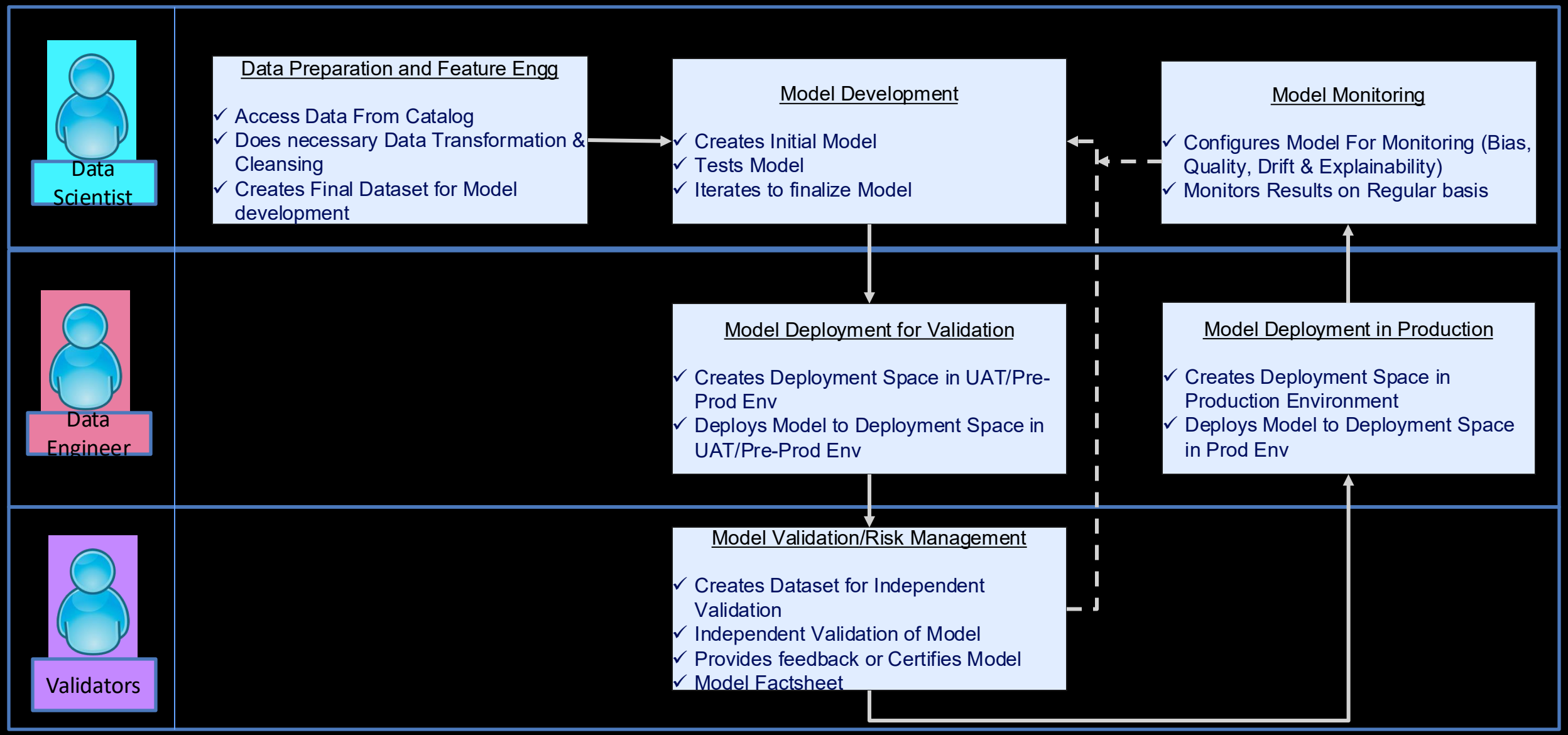
# ML Operationalization with IBM Cloud Pak For Data



# ML Ops In Action (1/2) - Data Provisioning and Governance

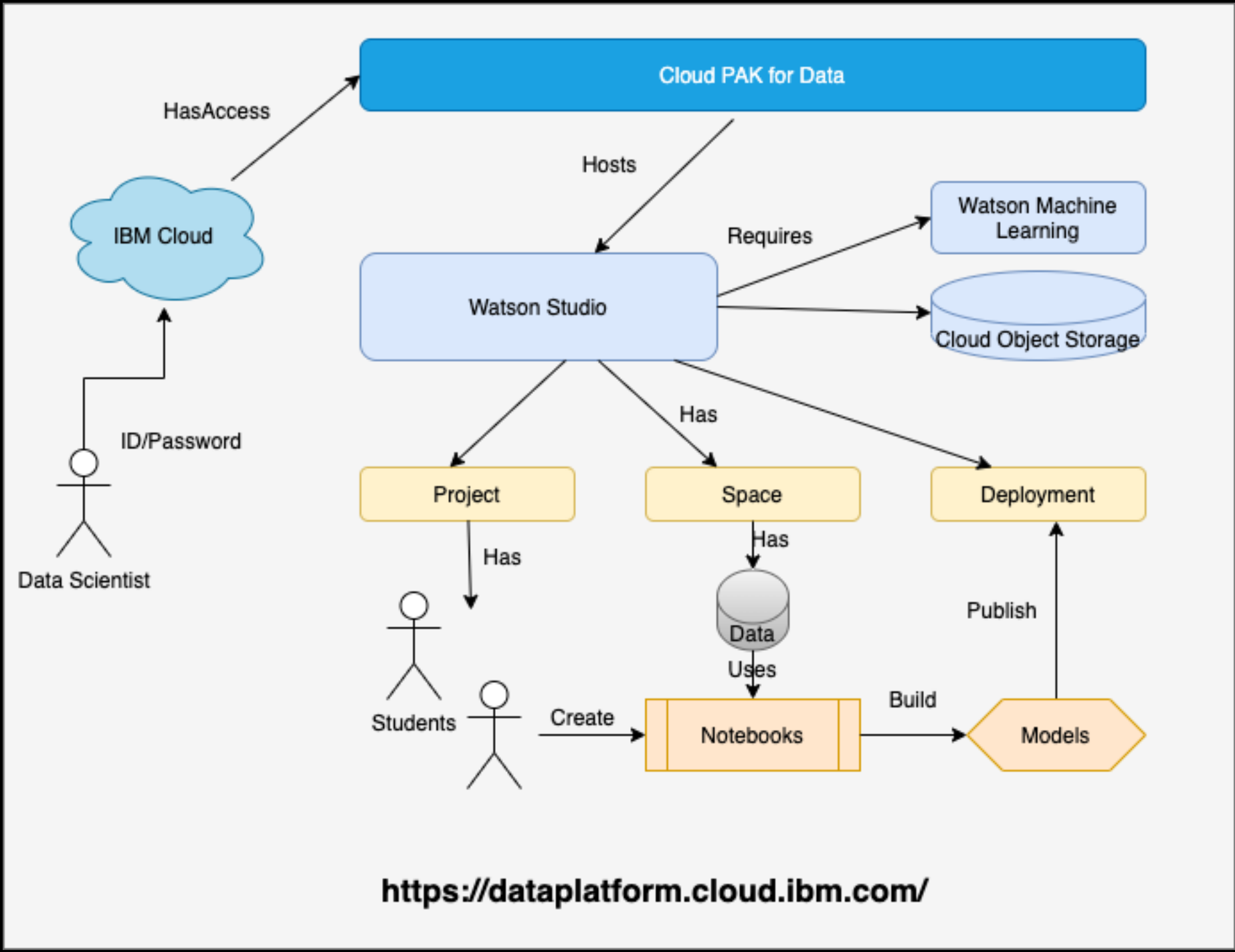


# ML Ops In Action (2/2) - Model Development, Deployment & Monitoring



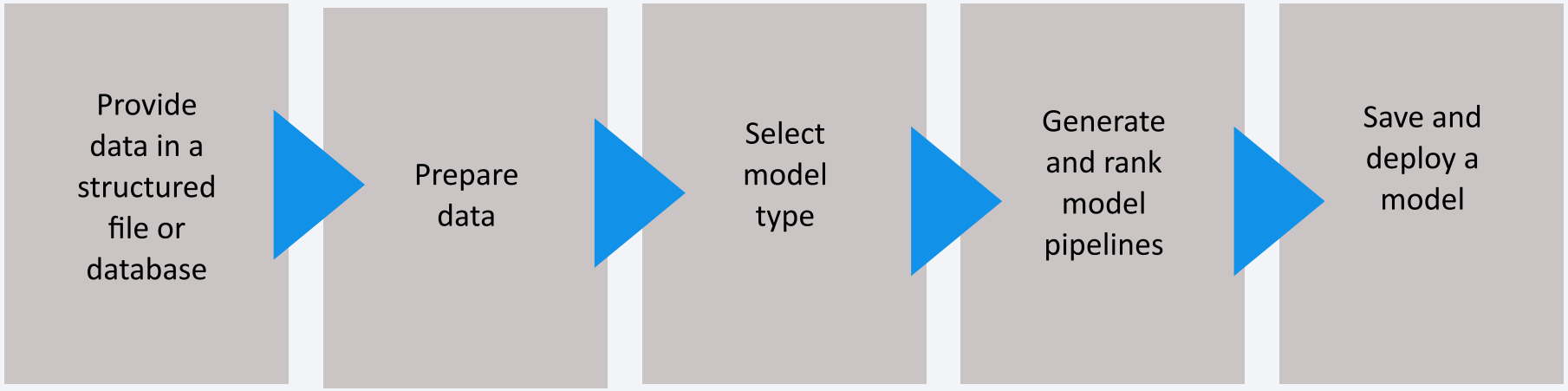
# Demos - Lab

# Watson Studio Model Deployment



# Watson Studio Model Deployment

## AutoAI




Feature type and detection

Missing values imputation

Feature encoding and scaling

Selection of best algorithm for the data

Hyper-parameter optimization

Optimized feature engineering

Ensemble for incremental learning

# Watson Studio Model Deployment

https://developer.ibm.com/learningpaths/get-started-watson-studio/

IBM | IBM **Developer**  Topics  Products & Services  Engagement  Open source at IBM

Data science > Analyze structured and unstructured data to extract knowledge and insights.

Learning Path

**Take control of your data with Watson Studio**

Save  Like

- **Overview**
- Introduction to IBM Watson Studio
- Initial data exploration
- Build, deploy, and trust models
- Test and validate machine learning models
- Summary

Learning Path

## Take control of your data with Watson Studio

Tutorials, articles, and other resources to solve your data science and machine learning needs

By Rich Hagarty, Einar Karlsen

Updated December 12, 2022 | Published June 1, 2021

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## Overview

IBM Watson Studio simplifies the process of experimentation to deployment, as well as data exploration, model development, and training. It lets you build, run, and manage AI models, and optimize decisions anywhere on IBM Cloud Pak for Data. You can bring together open source frameworks like PyTorch, TensorFlow, and scikit-learn with IBM for code-based and visual data

Your privacy choices

https://dataplatform.cloud.ibm.com/docs/content/wsj/analyze-data/autoai-overview.html?context=cpdaas

# Watson Studio Model Deployment

IBM Cloud Pak for Data | Search in your workspaces | Upgrade

Docs / Analyzing data and working with ... / AutoAI

Find information


- Overview
- Services and integrations
- Getting started
- Projects
- Preparing data
- Analyzing data and working with models
  - Notebooks and scripts
  - Cognos Dashboards
  - AutoAI
  - Federated learning
  - Decision Optimization
  - SPSS Modeler
  - Visualizing your data
- Deploying models and other assets
- Data governance
- AI governance
- Troubleshooting
- Administration
- Glossary


## AutoAI Overview

Last updated: Jan 12, 2024

The AutoAI graphical tool analyzes your data and uses data algorithms, transformations, and parameter settings to create the best predictive model. AutoAI displays various potential models as model candidate pipelines and rank them on a leaderboard for you to choose from.

Show tools & services map

**Required service**

Watson Machine Learning

Watson Studio

**Data format**

Tabular: CSV files, with comma (,) delimiter for all types of AutoAI experiments.

Connected data from IBM Cloud Object Storage.

> **Note:**
> You can use a data asset that is saved as a ***Feature Group (beta)*** but the metadata is not used to populate the AutoAI experiment settings.

**Data size**

Up to 1 GB or up to 20 GB. For details, refer to AutoAI data use.

## AutoAI data use

These limits are based on the default compute configuration of 8 CPU and 32 GB.

AutoAI classification and regression experiments:

- You can upload a file up to 1 GB for AutoAI experiments.

# Watson Studio Model Deployment

https://www.ibm.com/downloads/cas/XGVY8QA5

https://arxiv.org/pdf/2007.01977.pdf

https://mediacenter.ibm.com/media/IBM+Watson+Machine+LearningA+Run+a+sample+AutoAI+experiment/1_6fl7dh43/99375561

https://www.coursera.org/learn/ibm-rapid-prototyping-watson-studio-autoai#modules

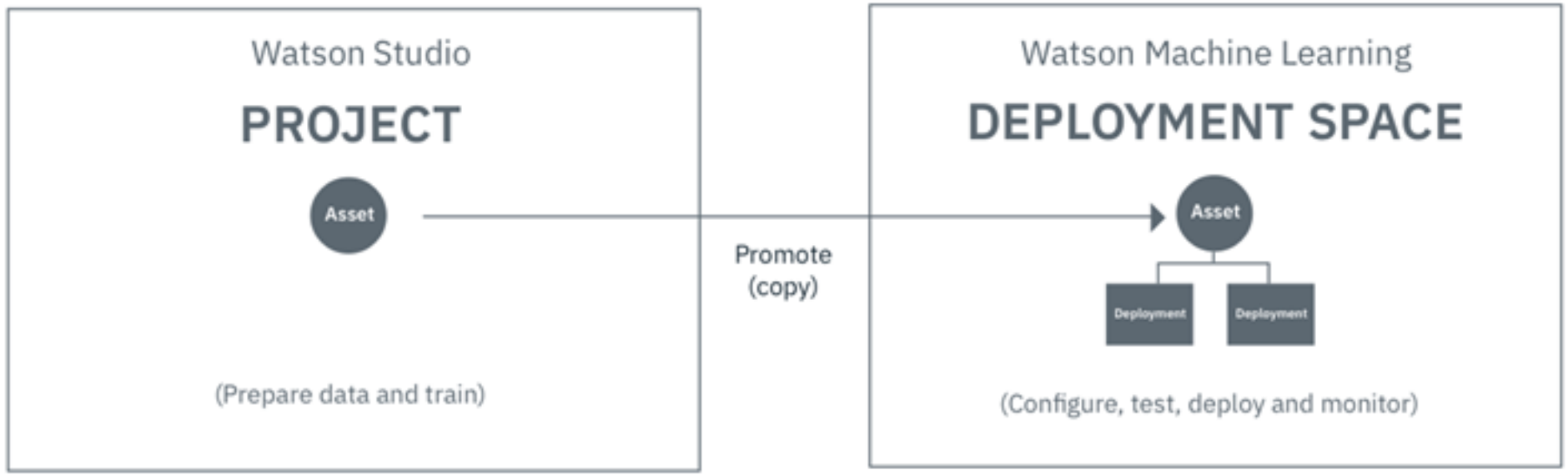
**Building a Rapid Prototype with Watson Studio AutoAI**
Module 1 • 2 hours to complete

**Automated Data Preparation and Model Selection**
Module 2 • 2 hours to complete

**Automated Feature Engineering and Hyperparameter Optimization**
Module 3 • 2 hours to complete

**Evaluation and Deployment of AutoAI-generated Solutions**
Module 4 • 2 hours to complete

# Watson Studio Model Deployment



# Deployment Options

# OBJECTIVES

- Explore different deployment options for TensorFlow models, focusing on practical implementations using a frozen protobuf (pb) model and labels.

https://ml-exchange.org/models/max-object-detector/

- Learn how to deploy the model as a Docker container, a Flask web application running on a virtual machine (VM), serverless functions, and other deployment options.

https://hub.docker.com/r/codait/max-object-detector

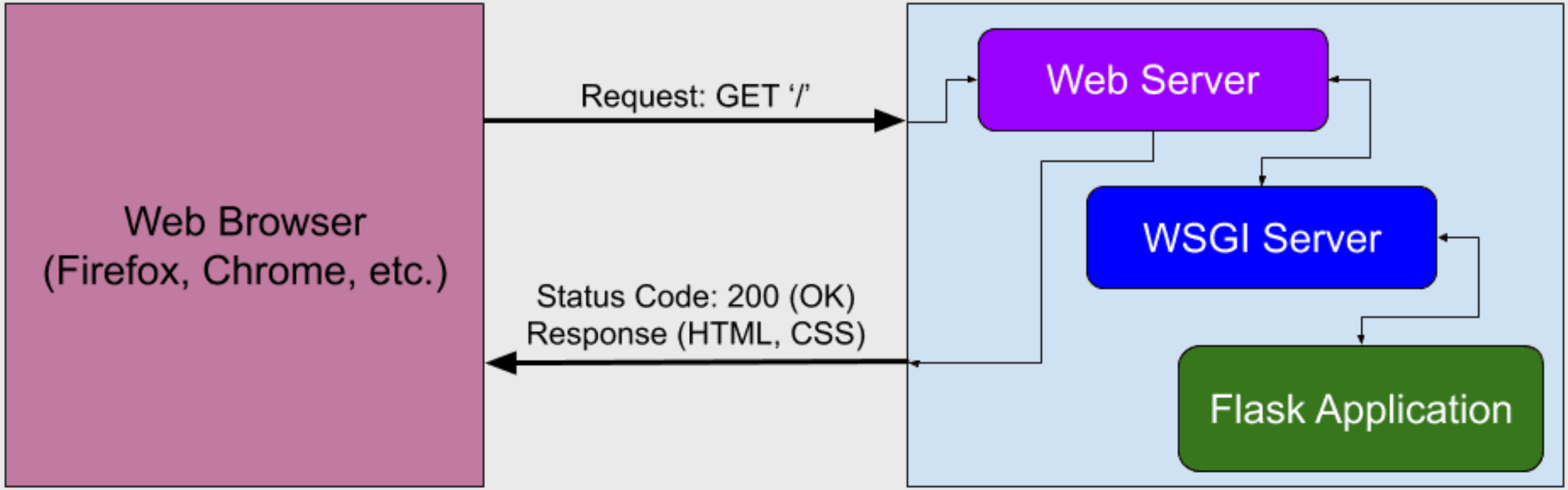
Machine Learning eXchange

# Deployment Options Overview

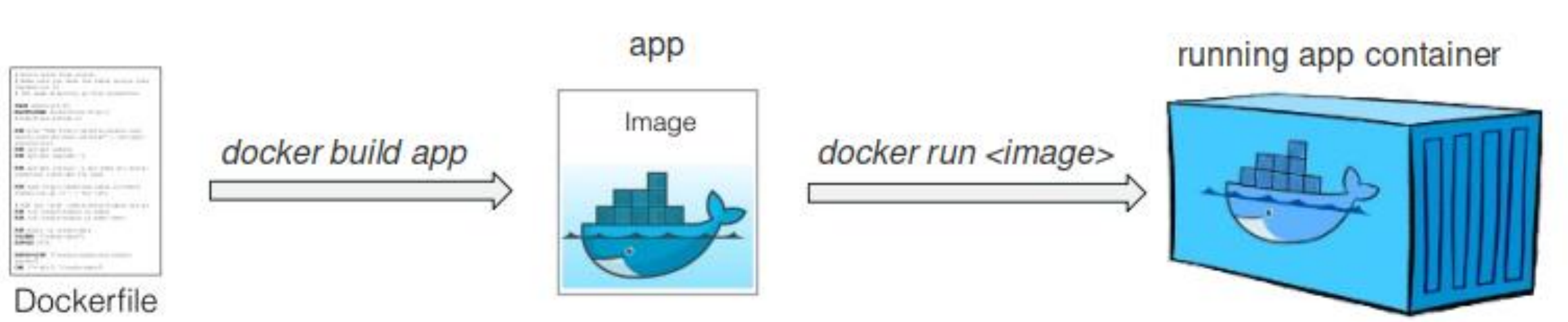
- Overview of different deployment options for TensorFlow models, including Docker containers, Flask web applications, serverless functions (e.g., AWS Lambda), and other cloud-based services.

- Pros and cons of each deployment option in terms of scalability, flexibility, ease of deployment, and cost.

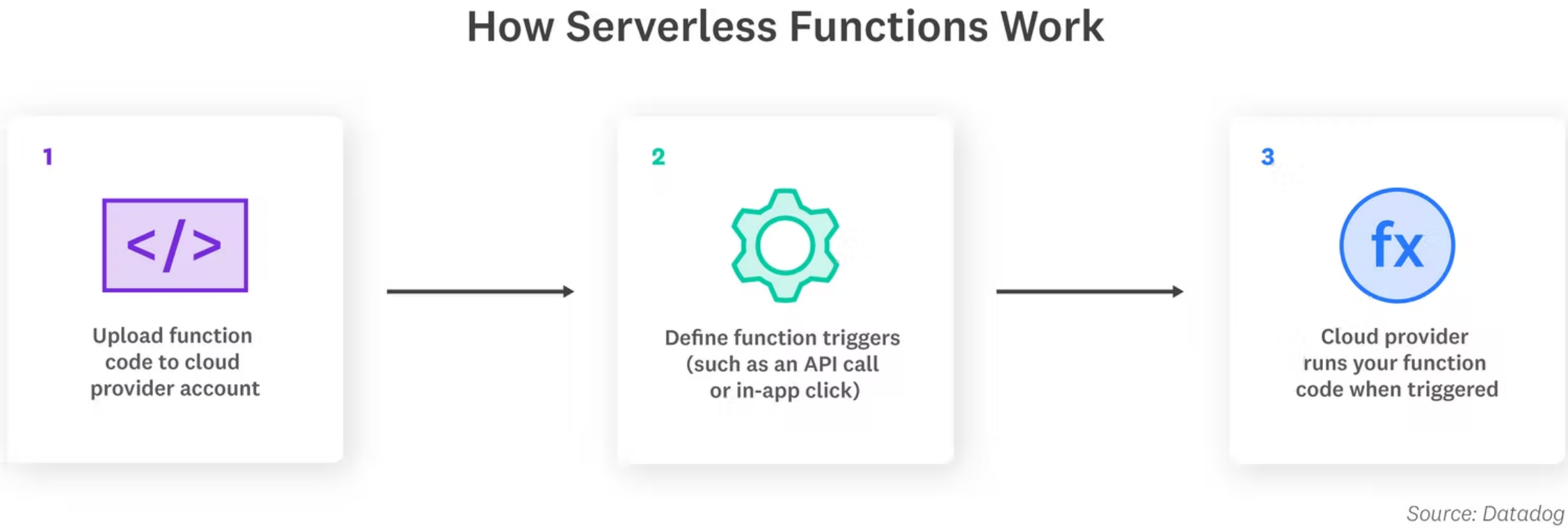
# Flask Web app (VM)



# As Docker Container



# Serverless Function



# Flask App (VM)

**Control**

Deploying a TensorFlow model as a Flask web application on a virtual machine provides greater control over the underlying infrastructure and environment.

**Customization**

Flask apps offer flexibility in configuring endpoints, middleware, and request handling logic, allowing for tailored deployment solutions.

**Integration**

Flask apps can easily integrate with existing web services, databases, and authentication systems, making them suitable for building complex applications.

**Security**

Running on a virtual machine provides isolation and security features, allowing administrators to enforce access controls, network policies, and security patches.

# Docker Pros & Cons

**Isolation**

Docker containers provide a lightweight and portable way to package applications and their dependencies, ensuring consistency and reproducibility across different environments.

**Scalability**

Containers can be easily scaled up or down to handle varying workloads, making them suitable for both development and production environments.

**Flexibility**

Docker containers allow for easy integration with other containerized services, microservices architectures, and container orchestration platforms like Kubernetes.

**Resource Efficiency**

Containers consume fewer resources compared to virtual machines, leading to better resource utilization and cost savings.

# Cloud Serverless Function

**Auto-Scaling**

Serverless functions automatically scale up or down based on incoming requests, ensuring optimal performance and resource utilization without manual intervention.

**Pay-per-Use Pricing**

Serverless platforms charge based on the actual execution time and resources consumed, resulting in cost savings for sporadic or unpredictable workloads.

**Simplified Operations**

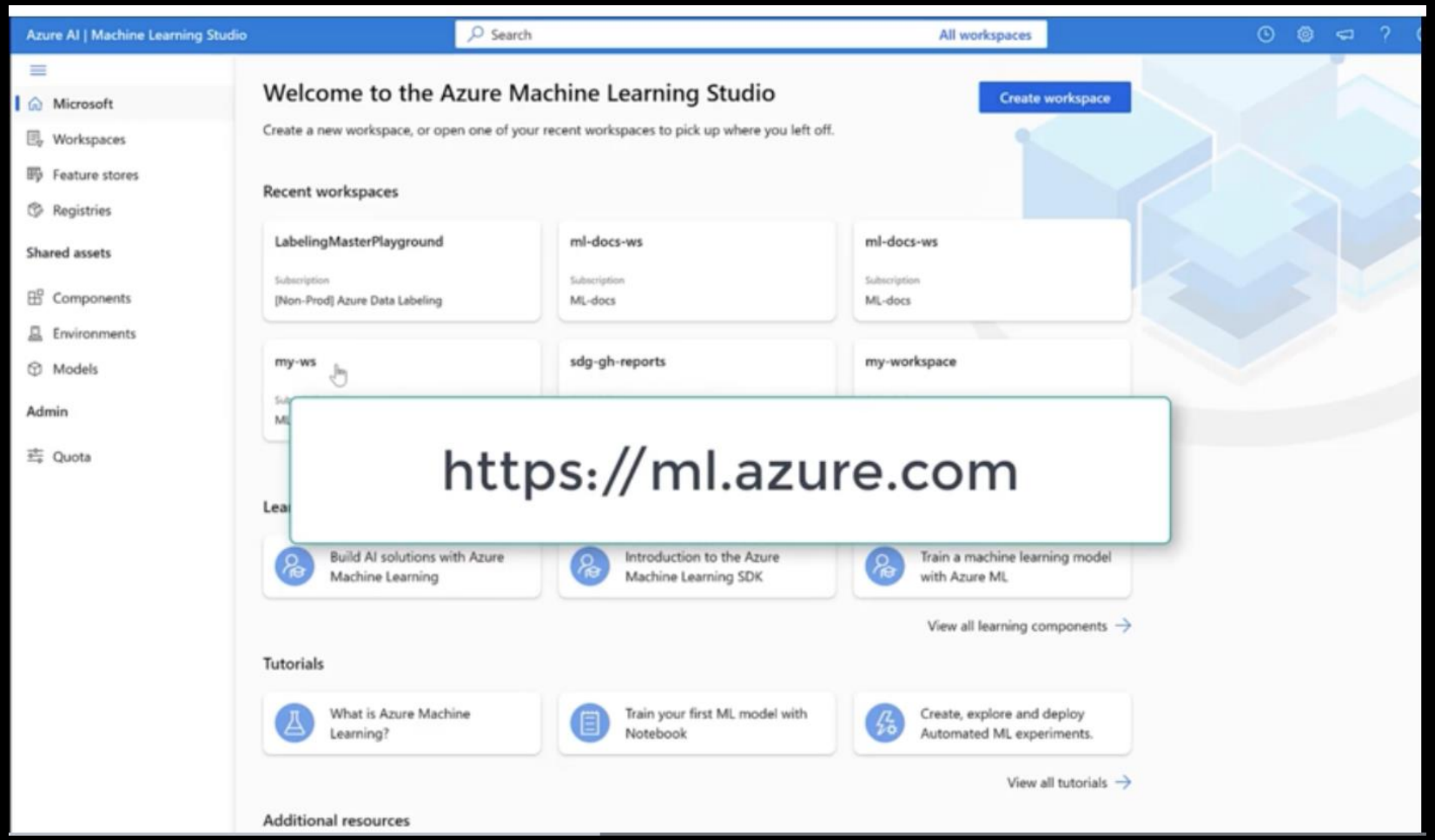
Serverless platforms abstract away infrastructure management tasks, such as provisioning, scaling, and monitoring, allowing developers to focus on application logic.

**Low Latency**

Serverless functions can be deployed closer to the end-user or data source, reducing network latency and improving response times for real-time applications.

# Deployments cont...

# Azure ML Studio




https://learn.microsoft.com/en-us/azure/machine-learning/tutorial-deploy-model?view=azureml-api-2

# Azure ML Studio

- The steps you take are:

- Register your model
- Create an endpoint and a first deployment
- Deploy a trial run
- Manually send test data to the deployment
- Get details of the deployment
- Create a second deployment
- Manually scale the second deployment
- Update allocation of production traffic between both deployments
- Get details of the second deployment
- Roll out the new deployment and delete the first one

https://learn.microsoft.com/en-us/azure/machine-learning/tutorial-deploy-model?view=azureml-api-2

# AI @ Edge

https://github.com/dusty-nv/jetson-inference

## Deploying Deep Learning

Welcome to our instructional guide for inference and realtime vision DNN library for **NVIDIA Jetson** devices. This project uses **TensorRT** to run optimized networks on GPUs from C++ or Python, and PyTorch for training models.

Supported DNN vision primitives include `imageNet` for image classification, `detectNet` for object detection, `segNet` for semantic segmentation, `poseNet` for pose estimation, and `actionNet` for action recognition. Examples are provided for streaming from live camera feeds, making webapps with WebRTC, and support for ROS/ROS2.

Image Classification

Object Detection

Semantic Segmentation

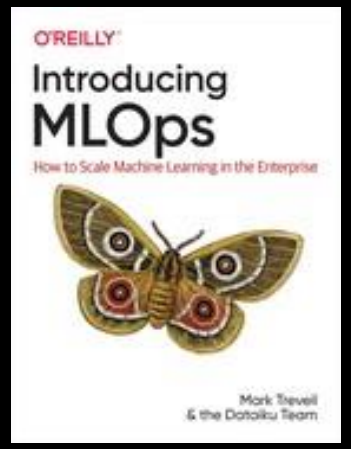
Pose Estimation

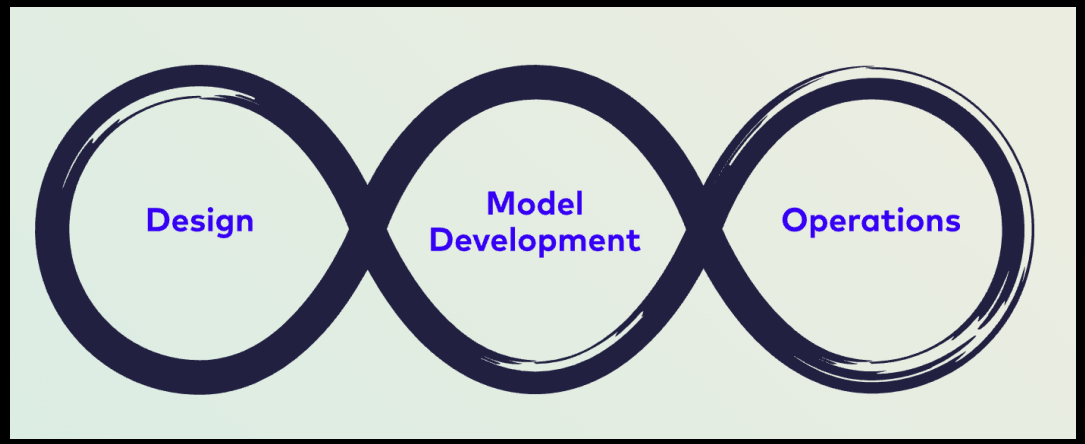
Action Recognition

Background Removal

Mono Depth

# Q&A



https://ml-ops.org/

https://aws.amazon.com/what-is/mlops

https://www.ibm.com/think/topics/mlops

https://medium.com/inside-machine-learning/ai-ops-managing-the-end-to-end-lifecycle-of-ai-3606a59591b0

https://learn.microsoft.com/en-us/azure/machine-learning/concept-model-management-and-deployment